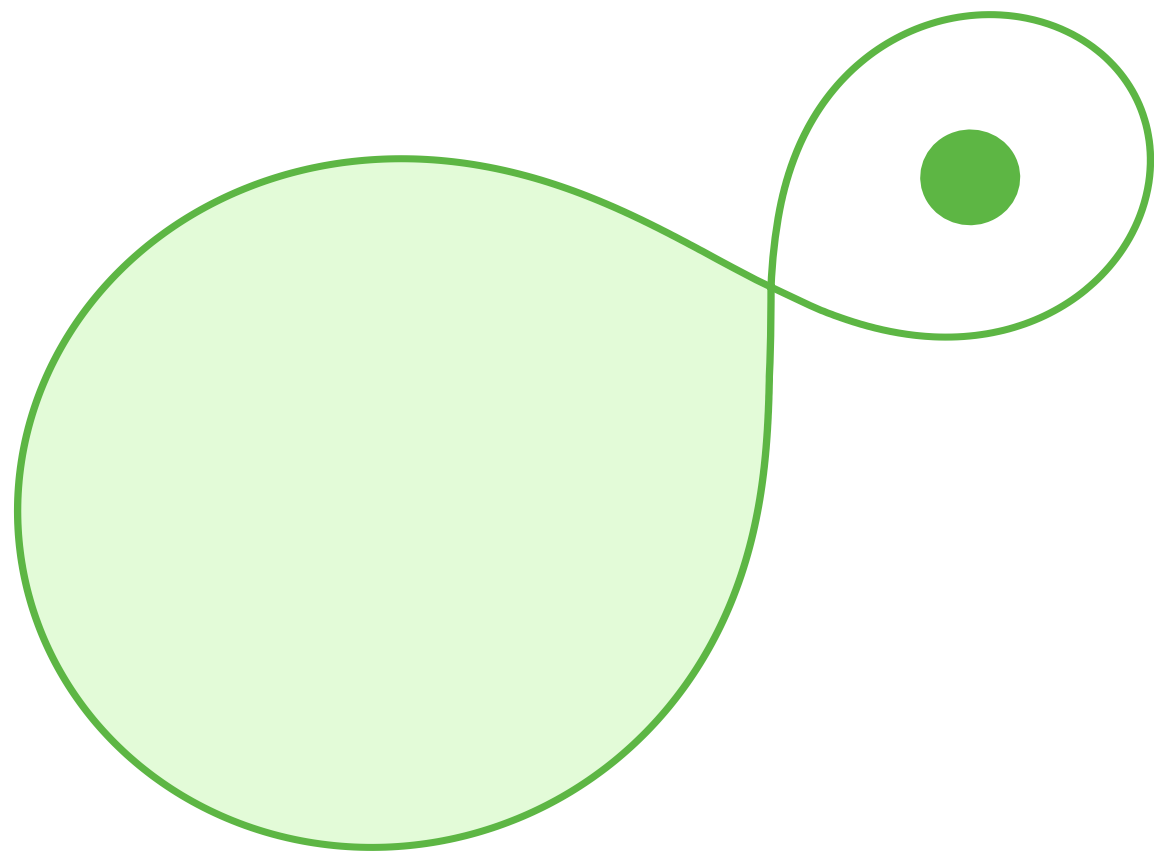


Forming barium stars through stable RLOF mass transfer from TPAGB-stars

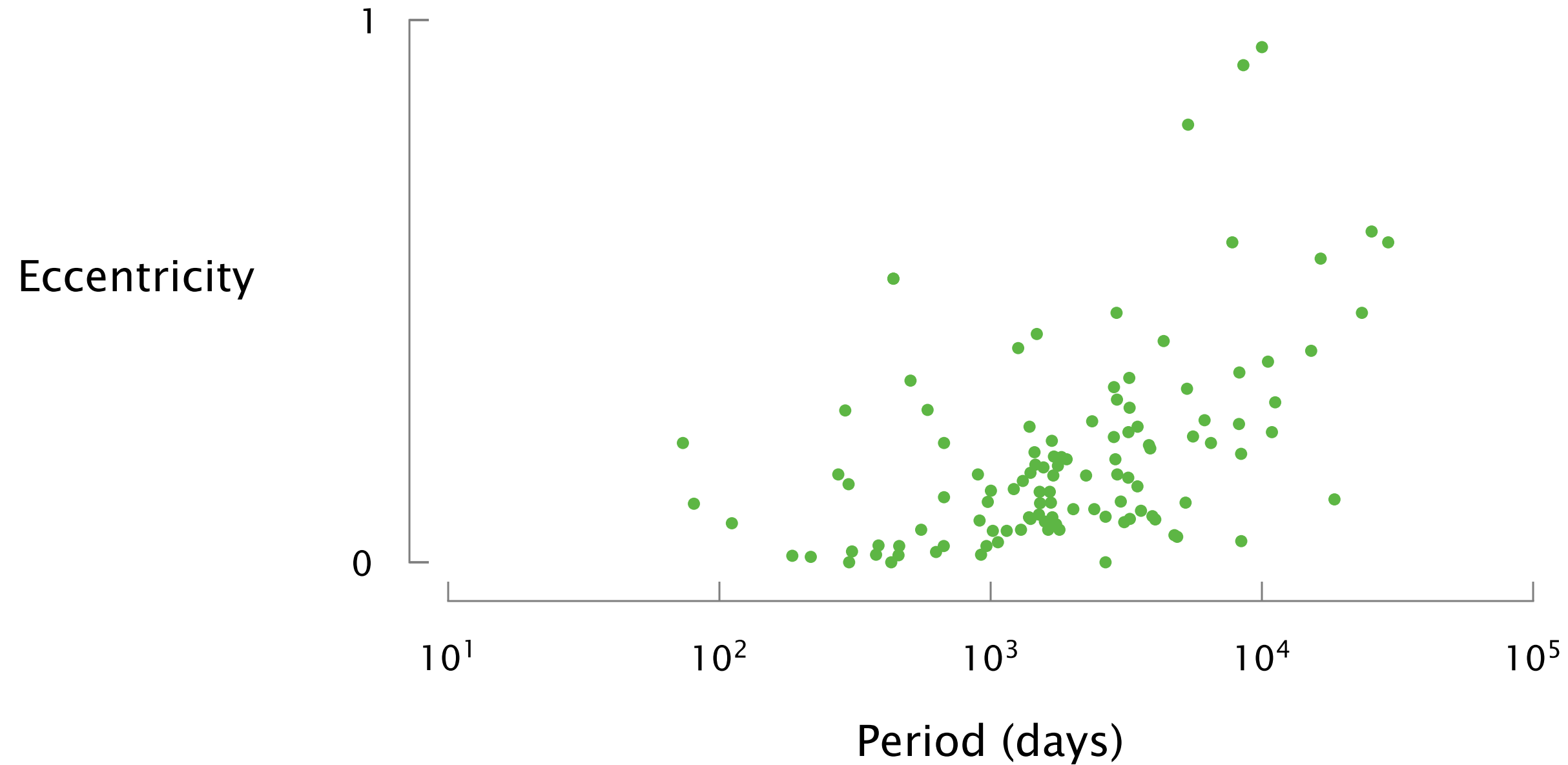


Koen Essers

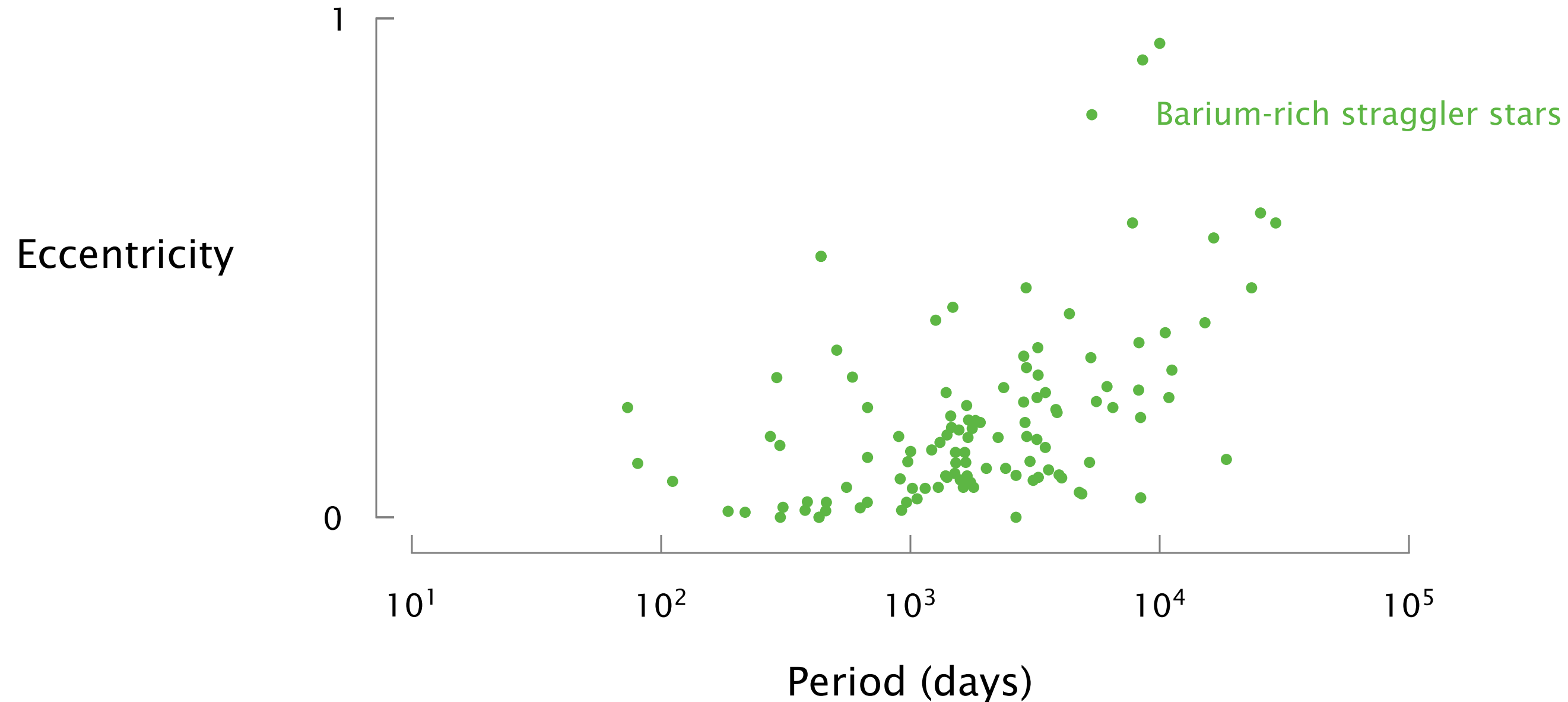
Radboud University

10th of June 2026

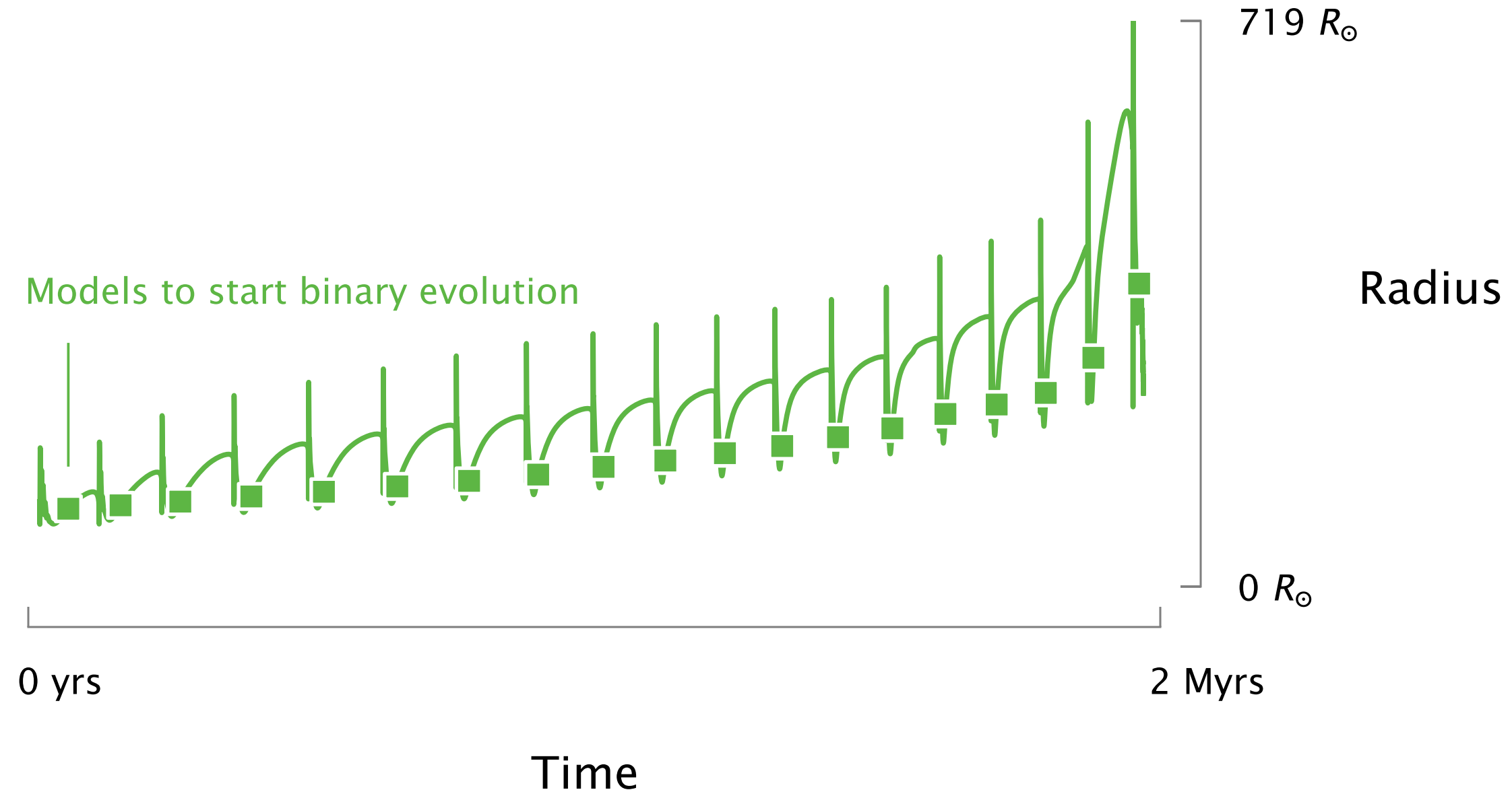
Post-mass-transfer binaries exist over a broad range of periods



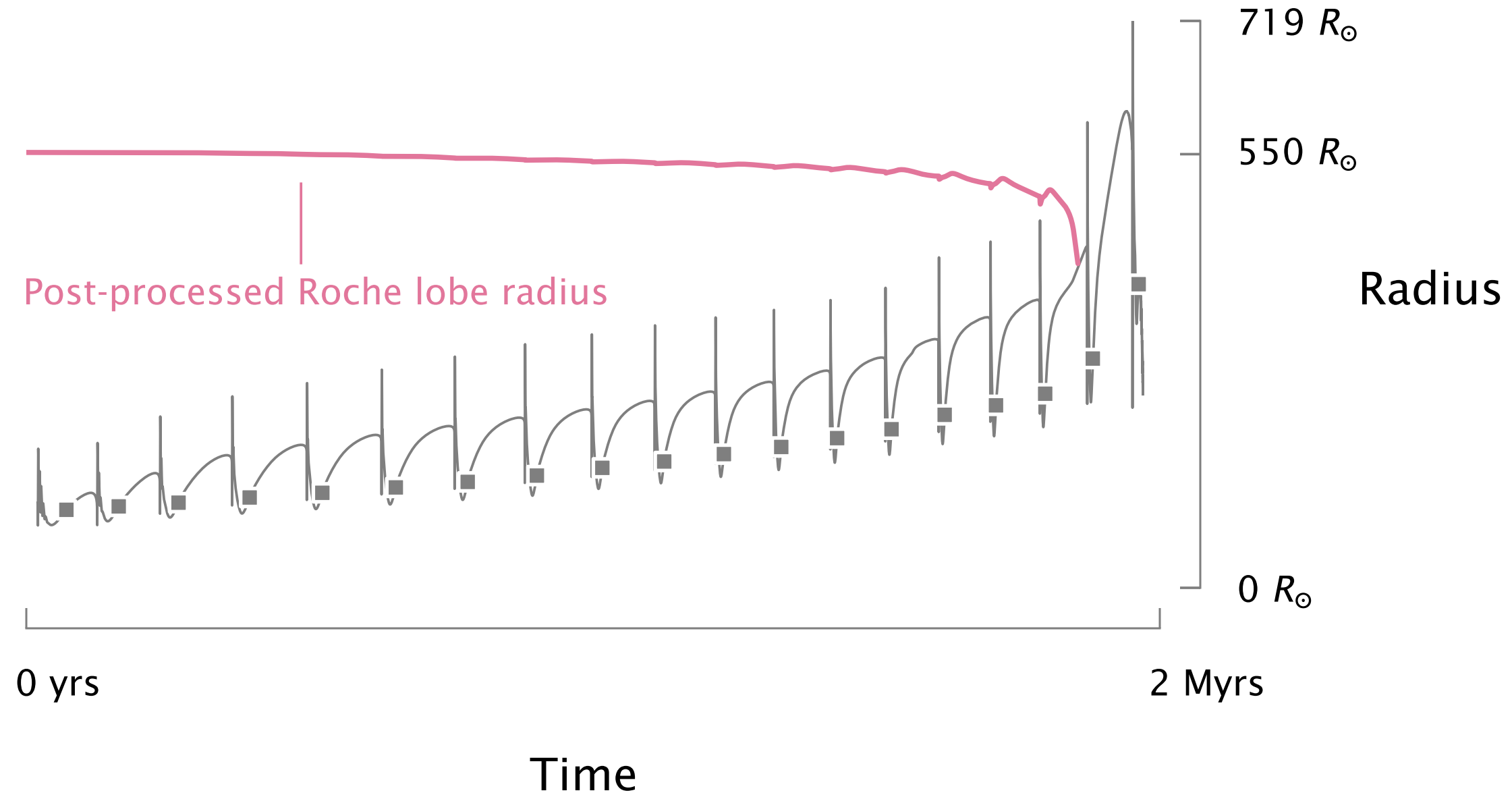
Ba stars have periods incompatible with wind mass transfer or common envelopes



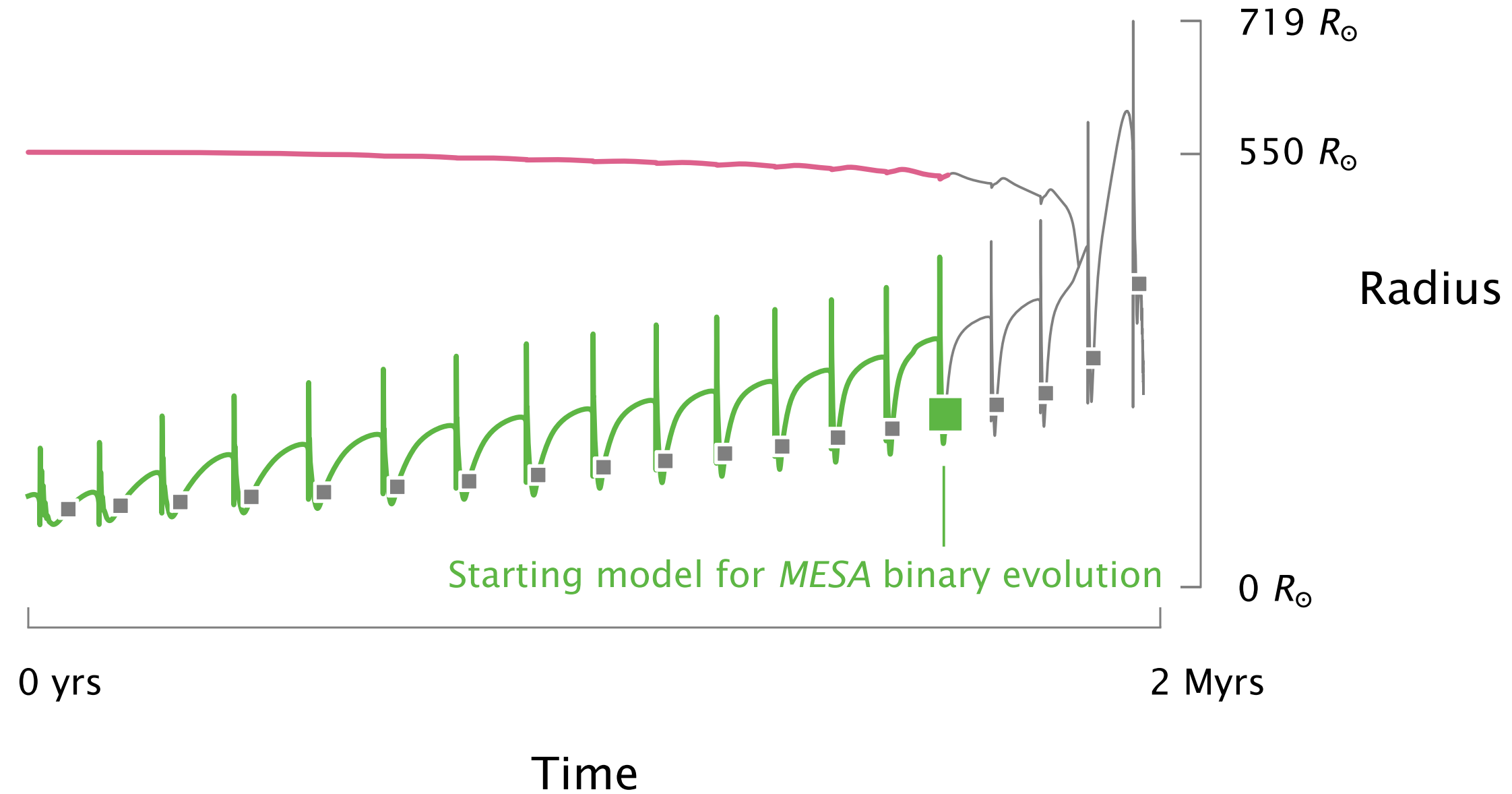
I simulate detailed TP-AGB stars with *MESA*
and I save models to use as binary input



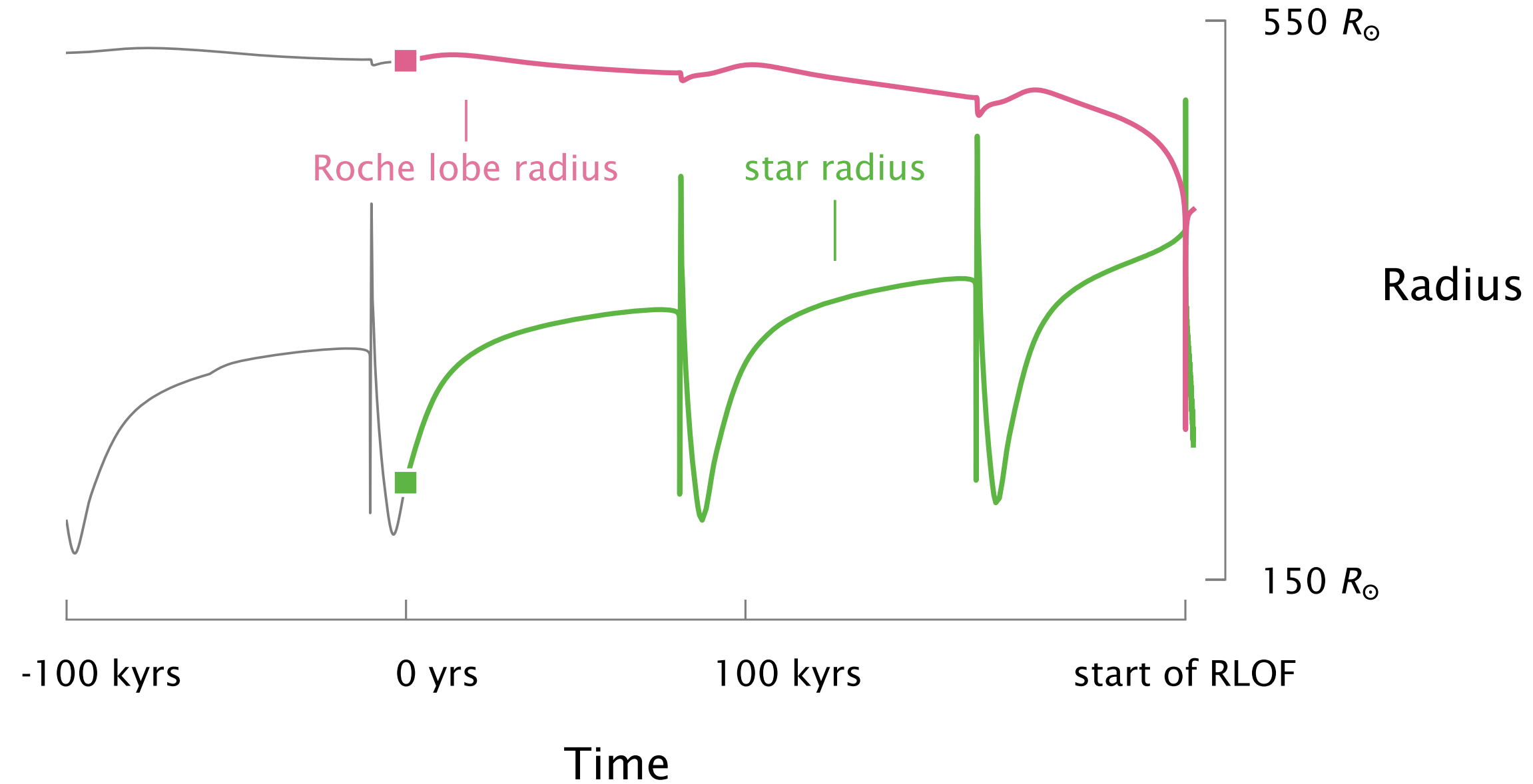
I predict how the binary evolves using single star models and a post-processing step



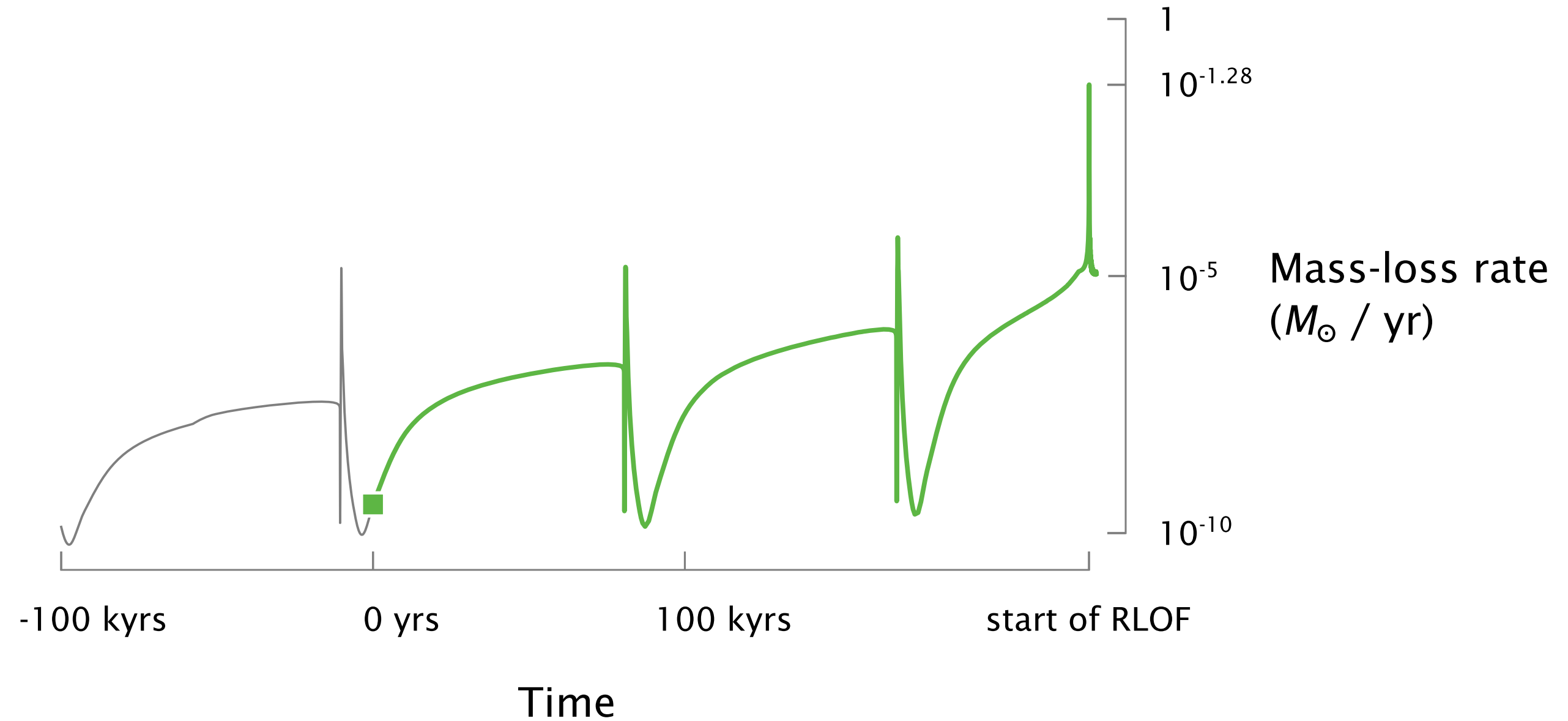
I determine a *MESA* starting model and
infer the rotation, separation and mass-ratio



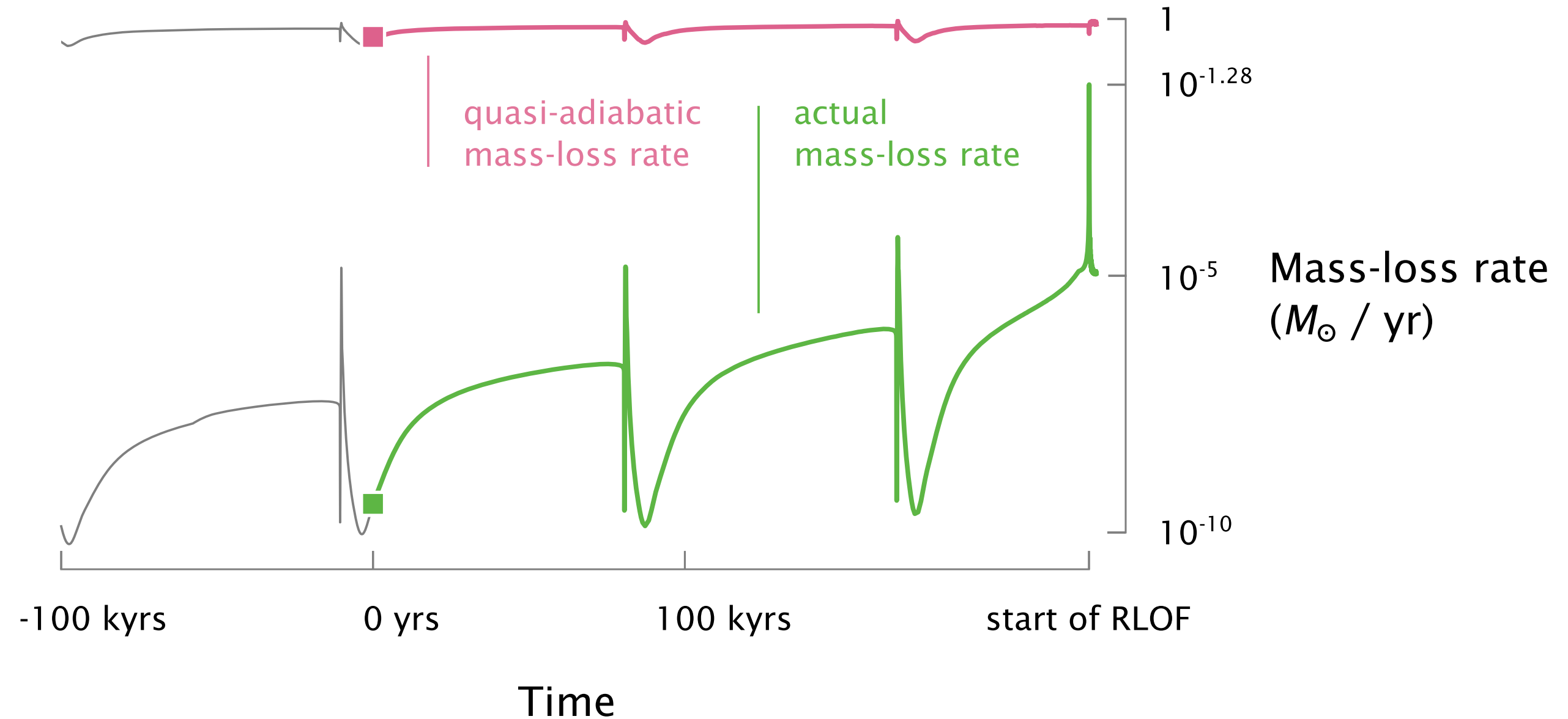
Tides, wind mass-transfer and stellar evolution drive systems into RLOF



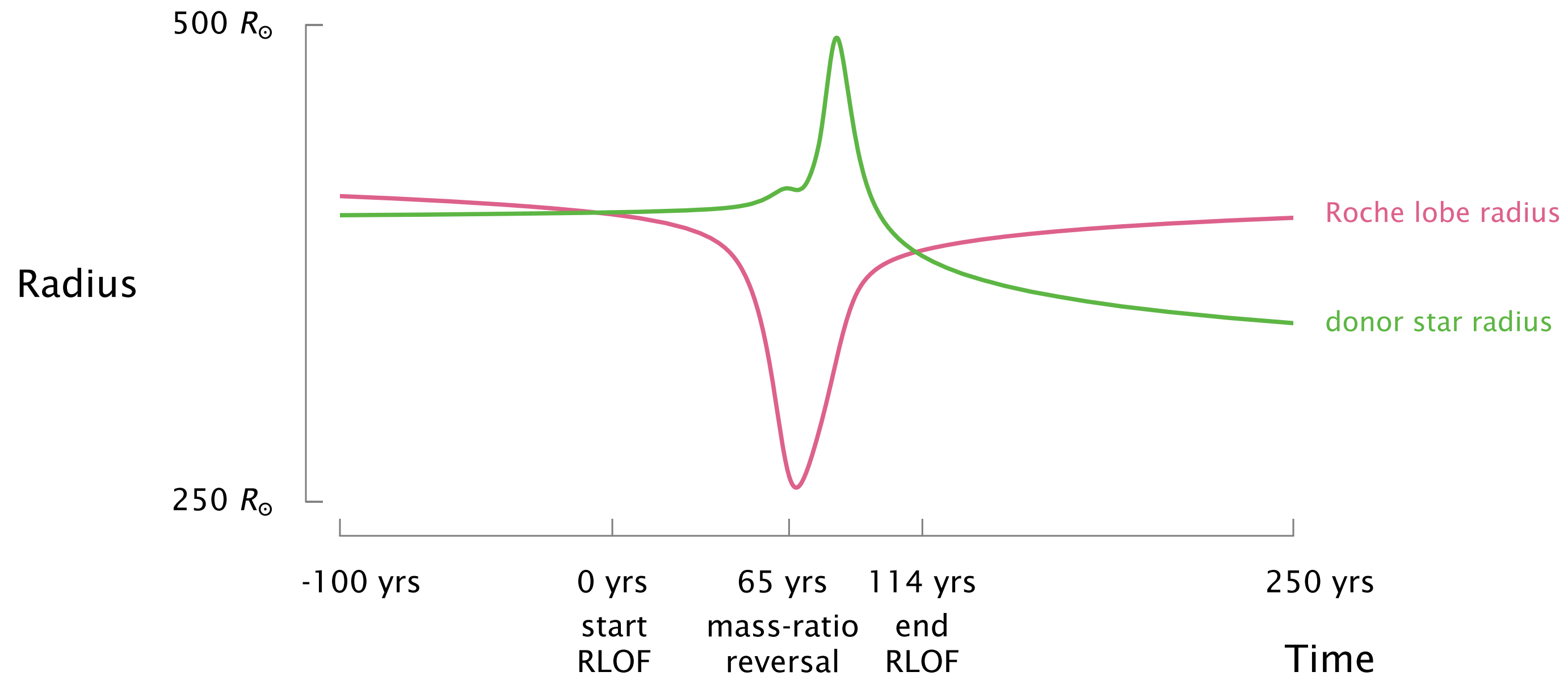
The donor experiences a short but strong burst of mass loss during RLOF



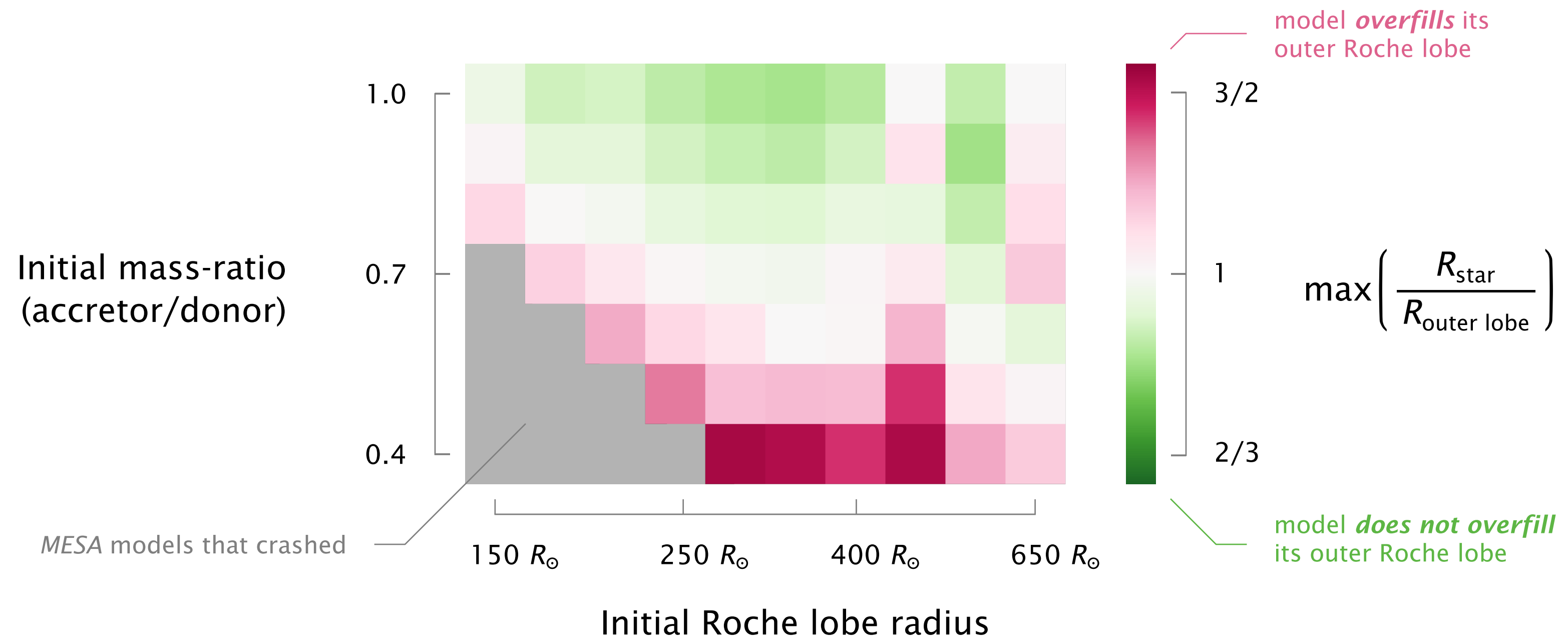
The quasi-adiabatic mass-loss rate is always larger than the actual mass loss rate

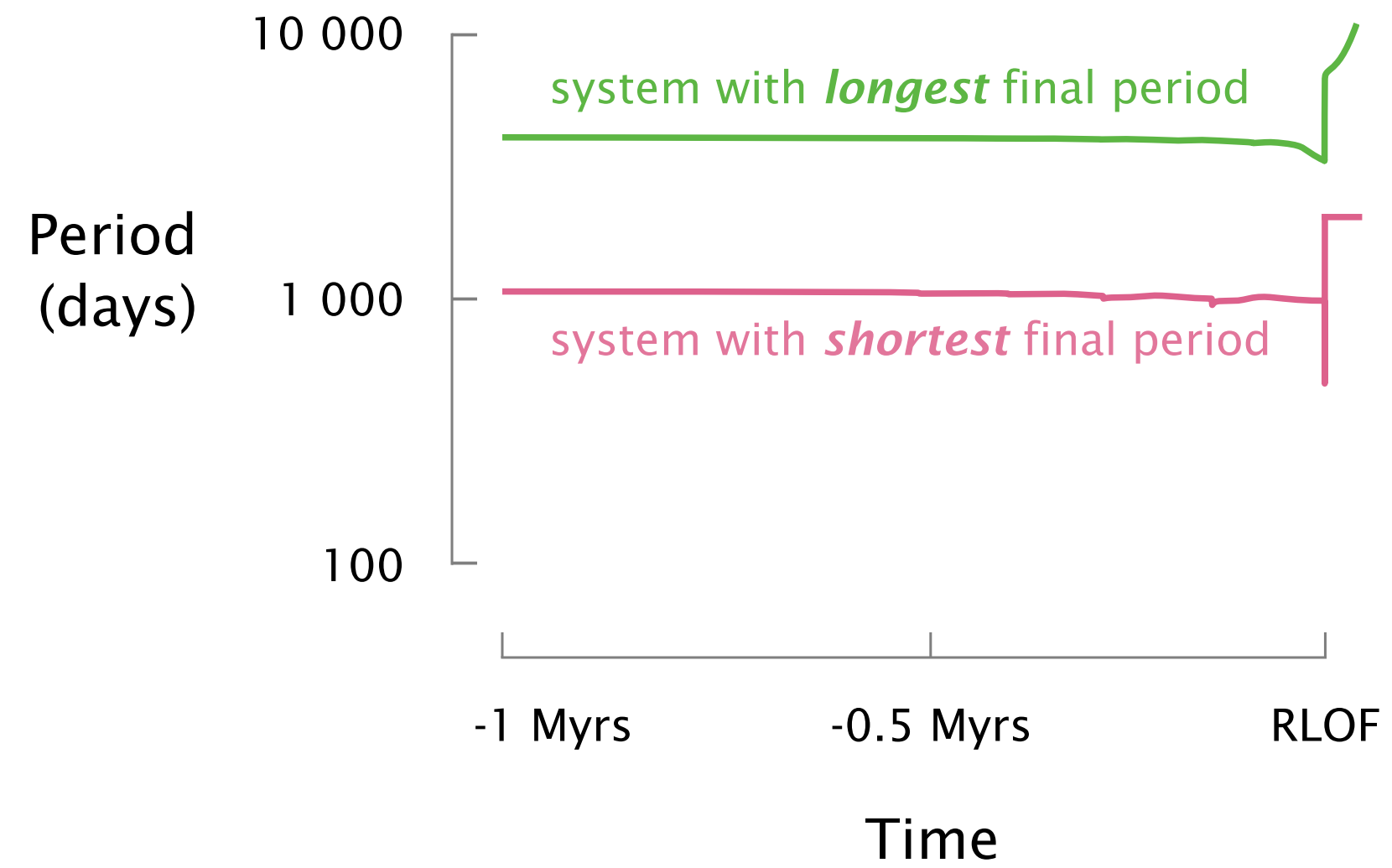


The donor significantly overfills its Roche lobe throughout the short RLOF episode

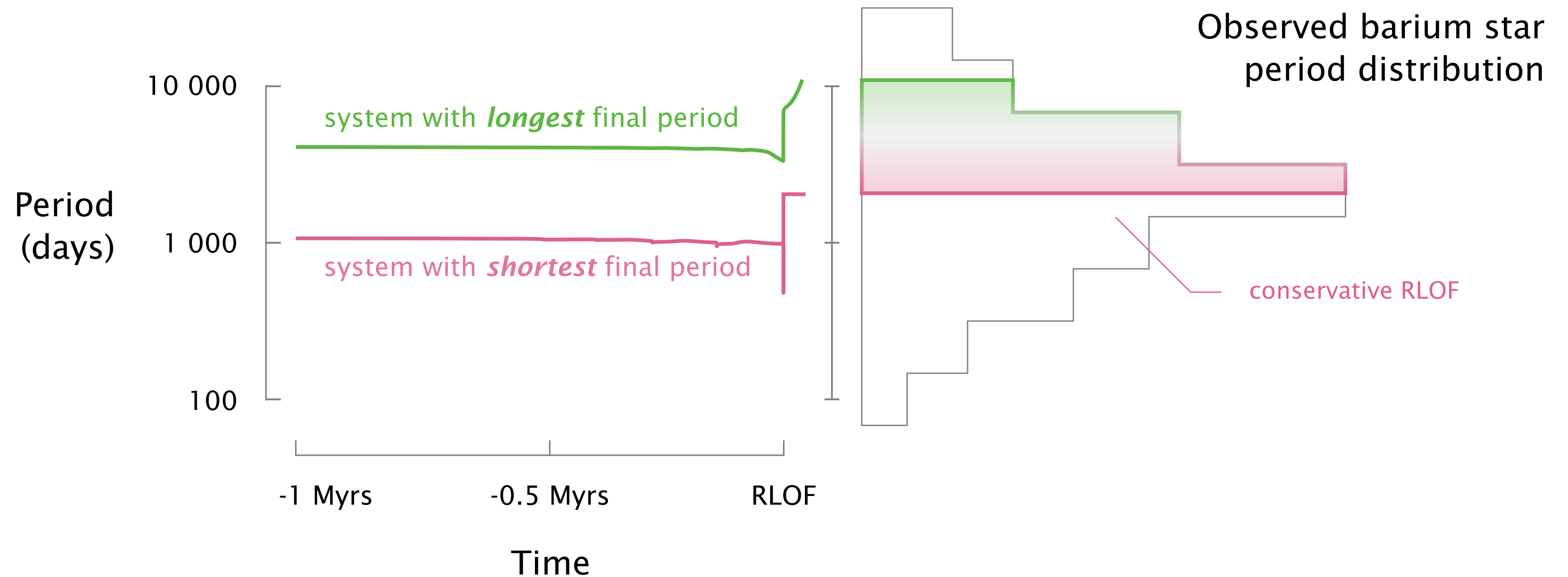


A significant fraction of models overfill their outer Roche lobe during mass-transfer

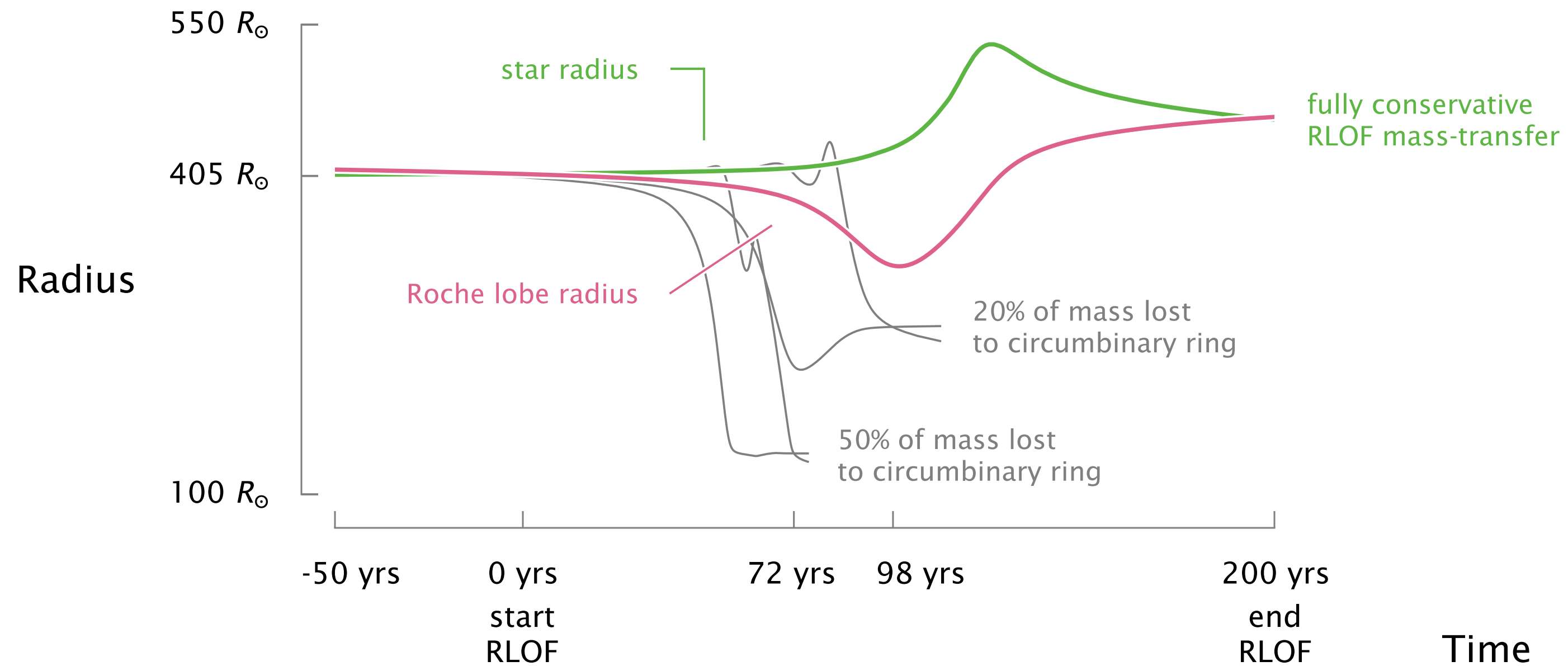




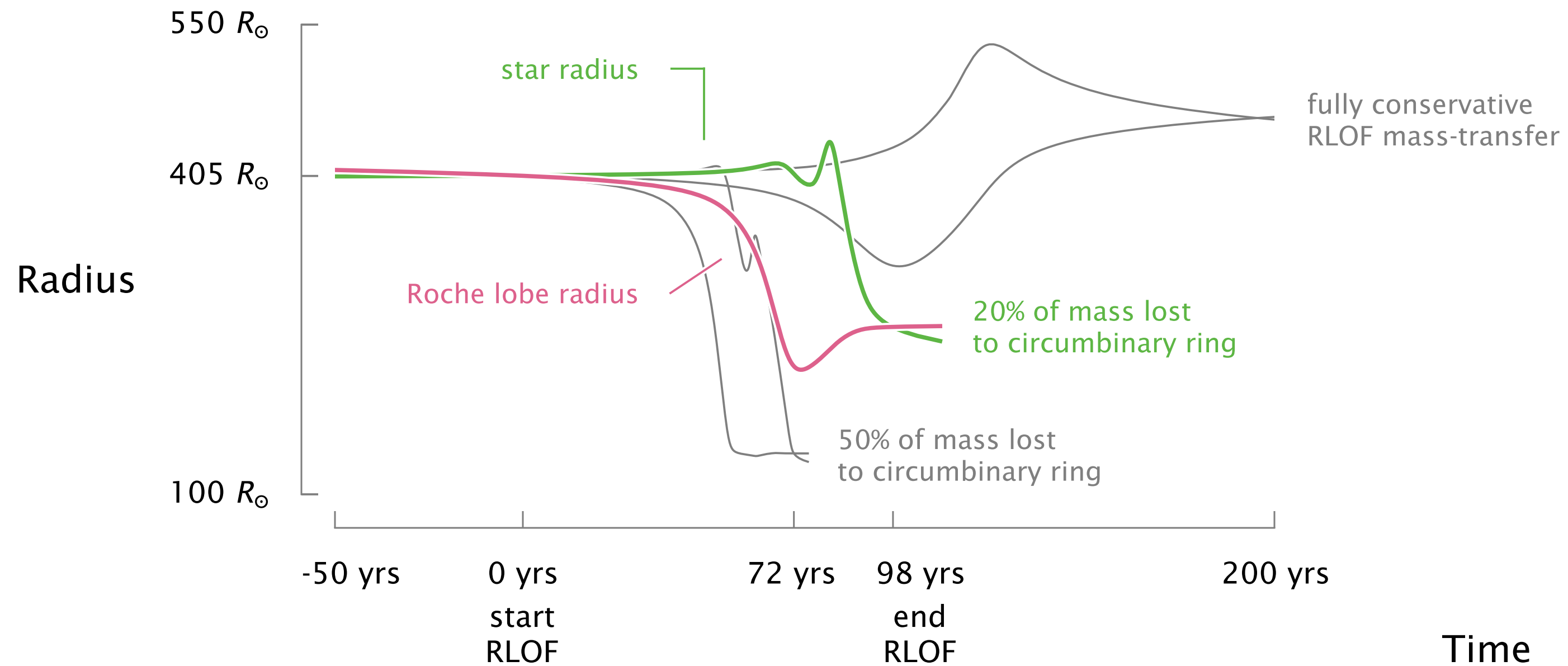
Conservative RLOF mass-transfer cannot explain observed systems with short periods



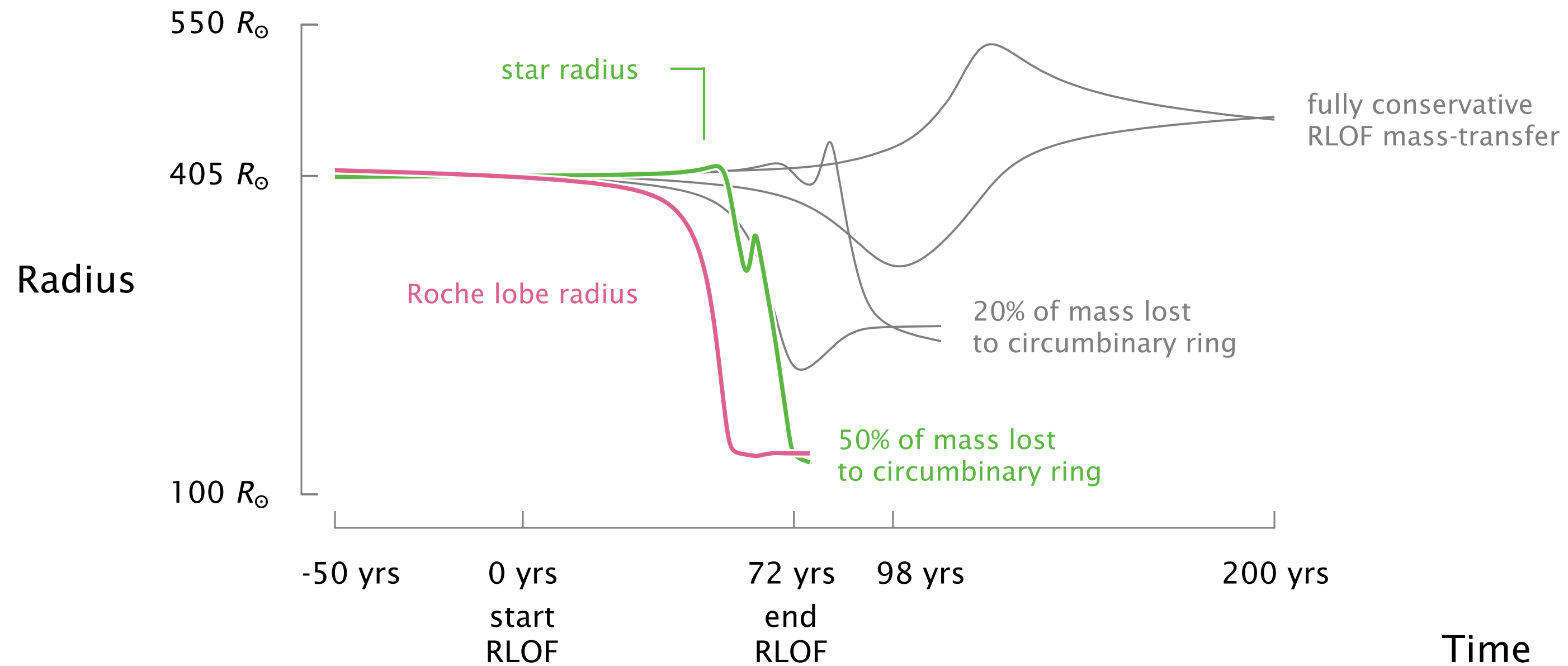
Non-conservative RLOF results in larger overfill ratios and shorter final periods



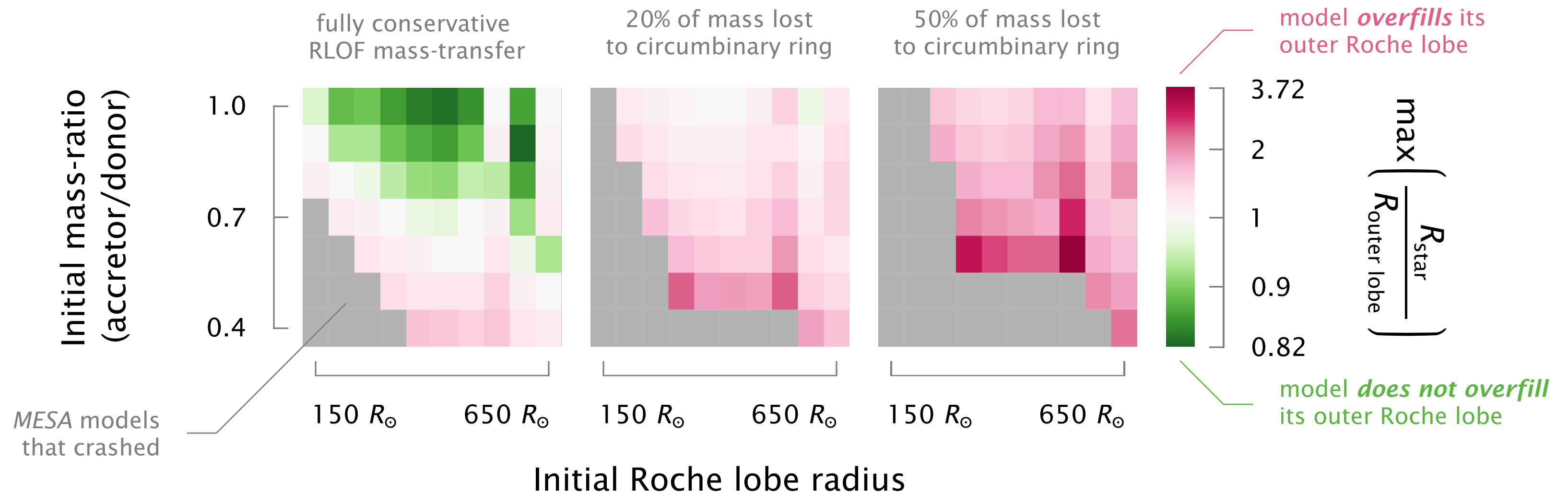
Non-conservative RLOF results in larger overflow ratios and shorter final periods



Non-conservative RLOF results in larger overfill ratios and shorter final periods



Mass-loss to a circumbinary ring enhances outer Roche lobe overfilling



Circumbinary mass-loss yields short periods, but still not the full observed distribution

